

TORCH syndrome.

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The original TORCH complex described clinically similar congenital infections caused by *Toxoplasma gondii*, rubella virus, cytomegalovirus, and herpes simplex virus, types 1 and 2. Cutaneous manifestations, including petechiae, purpura, jaundice, and dermal erythropoiesis, are commonly seen in toxoplasmosis, rubella, and cytomegalovirus infections. In herpes simplex virus infections, 80% of symptomatic infants show single or grouped cutaneous vesicles, oral ulcers, or conjunctivitis. Extracutaneous signs and symptoms are variable and can be severe. Significant clinical signs in congenital toxoplasmosis include diffuse intracerebral calcification, chorioretinitis, and microcephaly; congenital rubella can result in deafness, congenital heart disease, retinopathy, and brain calcification. Cytomegalic inclusion disease can include hepatomegaly, splenomegaly, paraventricular calcification, and intrauterine growth retardation. Localized or disseminated congenital herpes virus infection often involves the central nervous system and the eye. Diagnosis is confirmed by culture and identification of species-specific immunoglobulin M within the first 2 weeks of life. Histological examination contributes to the diagnosis in herpes simplex virus infection. Treatment for toxoplasmosis includes pyrimethamine with sulfadiazine or trisulfapyrimidine; congenital herpes simplex virus infection is treated with acyclovir. No specific therapy for congenital rubella or cytomegalovirus infections has been established, and so treatment is primarily supportive.

Nongenetic causes of hearing loss.

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Nongenetic as well as genetic etiologies must be explored in the child with identified hearing loss. Graduates of the neonatal intensive care unit are at increased risk for developing hearing loss due to hypoxia, hyperbilirubinemia, very low birth weight, and ototoxic medications. Although meningitis has decreased in frequency, it is still a risk factor for hearing loss. Cytomegalovirus remains the most common congenital infection and a relatively common etiology of hearing loss, which can be progressive. Preventable causes of hearing loss include those caused by head trauma, noise, and ototoxic medications. Identification of the etiology of hearing loss can facilitate the development of a treatment and management plan.

Cytomegalovirus DNA detection in Guthrie cards: role in the diagnostic work-up of childhood hearing loss.

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Cytomegalovirus (CMV) infection is the leading cause of congenital nongenetic sensorineural hearing loss (SNHL) and a major cause of prelingual SNHL that is not present at birth. Polymerase chain reaction (PCR) analysis of dried blood samples on the Guthrie card has been proposed as a sensitive and specific method to screen for congenital CMV infection. Prospectively, consecutive infants who failed universal neonatal hearing screening and children referred for a noncongenital SNHL (NCHL) were included and underwent a standard audiometric and etiologic work-up. DNA was extracted from dried blood spots on neonatal Guthrie cards and amplified by real-time PCR. Data were available for 96 cases. Mean age of the

universal neonatal hearing screening group was 3.8 +/- 2.4 months (n = 41). Auditory brain stem response thresholds were 72.9 +/- 20.2 dB nHL. A CMV-positive PCR was obtained in 4 babies. One test was considered false-positive. This resulted in a 7.3% prevalence of congenital CMV infections. Mean age of the NCHL group was 4.9 +/- 3.2 years (n = 55). Hearing loss was moderate in 37, severe in 5, and profound in 13 children. A CMV-positive PCR was obtained in 4 children (7.3%). Other causes of SNHL were excluded in the PCR positive cases of both study groups. We advocate PCR for CMV DNA detection on Guthrie cards in the etiologic work-up of childhood SNHL and recommend serologic confirmation to exclude false-positive PCR results. 7.3% of SNHL in babies with congenital hearing loss and children with NCHL could be attributed with this technique to congenital CMV infection.

Exploring Biomarkers in Nonalcoholic Fatty Liver Disease Among Individuals With Type 2 Diabetes Mellitus.

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Integrating biomarkers into a comprehensive strategy is crucial for precise patient management, especially considering the significant healthcare costs associated with diseases. Current studies emphasize the urgent need for a paradigm shift in conceptualizing nonalcoholic fatty liver disease (NAFLD), now renamed metabolic dysfunction-associated steatotic liver disease (MASLD). Biomarkers are emerging as indispensable tools for accurate diagnosis, risk stratification, and monitoring disease progression. This review classifies biomarkers into conventional and novel categories, such as lipids, insulin resistance, hepatic function, and cutting-edge imaging/omics, and evaluates their potential to transform the approach to MASLD among individuals with type 2 diabetes mellitus (T2D). It focuses on the critical role of biomarkers in early MASLD detection, enhancing predictive accuracy, and discerning responses to interventions (pharmacological or lifestyle modifications). Amid this discussion, the complexities of the relationship between T2D and MASLD are explored, considering factors like age, gender, genetics, ethnicity, and socioeconomic background. Biomarkers enhance the effectiveness of interventions and support global initiatives to reduce the burden of MASLD, thereby improving public health outcomes. This review recognizes the promising potential of biomarkers for diagnostic precision while candidly addressing the challenges in implementing these advancements in clinical practice. The transformative role of biomarkers emerges as a central theme, promising to reshape our understanding of disease trajectories, prognosis, and the customization of personalized therapeutic strategies for improved patient outcomes. From a future perspective, identifying early-stage biomarkers, understanding environmental impact through exposomes, and applying a multiomics approach may reveal additional insight into MASLD development.

Metabolic dysfunction-associated steatotic liver disease: Evolution of the final terminology.

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The medical term nonalcoholic fatty liver disease (NAFLD) was coined in 1986 for a condition that has since become the most prevalent liver disorder worldwide. In the last 3 years, the global professional community launched 2 consecutive efforts to purge NAFLD from the medical dictionary and recommended new terms based on disease pathophysiology rather than distinction from similar

conditions featuring liver steatosis. A consensus by renowned clinical scholars primarily residing in the Asian-Pacific region introduced metabolic dysfunction-associated fatty liver disease (MAFLD) as a new name to replace NAFLD in 2020. In 2023, a nomenclature and classification resulting in the term metabolic dysfunction-associated steatotic liver disease (MASLD) was developed by a large expert panel under the auspices of leading liver societies from Europe and Americas. These marked and rapid shifts in nomenclature have garnered the attention of many researchers and clinicians across the globe due to the multilevel impact of a frequent and potentially progressive chronic liver disease in both adult and pediatric populations. The proposed terminologies differ in several ways but they have more in common than differences. They both capture key features of liver disease associated with cardiometabolic risk factors and with significant impact on all-cause and liver-related mortality. The framework of MASLD has incorporated many innovative aspects of MAFLD and while several conceptual disparities remain a work in progress, global efforts should focus on new insights into disease pathogenesis, outcome trajectories, prevention, and treatment. Here, some of these challenges are discussed to facilitate this process.

Steatotic liver disease, MASLD and risk of chronic kidney disease.

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With the rising tide of fatty liver disease related to metabolic dysfunction worldwide, the association of this common liver disease with chronic kidney disease (CKD) has become increasingly evident. In 2020, the more inclusive term metabolic dysfunction-associated fatty liver disease (MAFLD) was proposed to replace the old term non-alcoholic fatty liver disease (NAFLD). In 2023, a modified Delphi process was led by three large pan-national liver associations. There was consensus to change the fatty liver disease nomenclature and definition to include the presence of at least one of five common cardiometabolic risk factors as diagnostic criteria. The name chosen to replace NAFLD was metabolic dysfunction-associated steatotic liver disease (MASLD). The change of nomenclature from NAFLD to MAFLD and then MASLD has resulted in a reappraisal of the epidemiological trends and associations with the risk of developing CKD. The observed association between MAFLD/MASLD and CKD and our understanding that CKD can be an epiphenomenon linked to underlying metabolic dysfunction support the notion that individuals with MASLD are at substantially higher risk of incident CKD than those without MASLD. This narrative review provides an overview of the literature on (a) the evolution of criteria for diagnosing this highly prevalent metabolic liver disease, (b) the epidemiological evidence linking MASLD to the risk of CKD, (c) the underlying mechanisms by which MASLD (and factors strongly linked with MASLD) may increase the risk of developing CKD, and (d) the potential drug treatments that may benefit both MASLD and CKD.

Critical appraisal of metabolic dysfunction-associated steatotic liver disease: Implication of Janus-faced modernity.

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The existing term non-alcoholic fatty liver disease (NAFLD) has raised substantial concerns due to its inherent disadvantages of using exclusionary diagnostic criteria and the stigmatizing word 'fatty.' Three pan-national liver associations set out to explore a new nomenclature to replace both NAFLD and its suggested alternative, metabolic (dysfunction)-associated fatty liver disease (MAFLD). They surveyed if a change in nomenclature and/or definition is favored and which nomenclature best communicates disease

characteristics and increases awareness. In lieu of NAFLD/MAFLD, metabolic dysfunction-associated steatotic liver disease (MASLD) has been chosen, and an umbrella term, steatotic liver disease (SLD), encompassing the whole spectrum of liver disease, has been proposed. It has been suggested that cardiometabolic risk factors should be considered when categorizing SLD patients. Furthermore, a new subcategory, MASLD with increased alcohol intake (MetALD), casts light on a neglected group of patients with moderate or more alcohol consumption. The importance of metabolic dysfunction was acknowledged in this new nomenclature, but the precise contribution of metabolic dysfunction and alcohol consumption to the development and progression of SLD remains unclear. Herein, we review hepatologists' and endocrinologists' perspectives on the new nomenclature, along with its possible impact on clinical practice. Although it is premature to predict the settlement of the new nomenclature, this review may help build more evidence for a soft landing of it in the future.

Evolving liver disease insights from NAFLD to MASLD.

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The recent renaming of 'non-alcoholic fatty liver disease' (NAFLD) to 'metabolic dysfunction-associated steatotic liver disease' (MASLD) emphasizes metabolic dysfunction in steatotic liver disease and advocates for tailored, comprehensive treatment strategies, driving forward the development of personalized care and innovative therapeutic approaches.

NAFLD No More: A Review of Current Guidelines in the Diagnosis and Evaluation of Metabolic Dysfunction-Associated Steatotic Liver Disease (MASLD).

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Provide a concise update on metabolic dysfunction-associated steatotic liver disease (MASLD), formerly known as non-alcoholic fatty liver disease (NAFLD), as well as a practical approach to screening and initial evaluation. Nomenclature changes have placed a greater focus on cardiometabolic risk factors in the definition of MASLD. Screening for MASLD is by stepwise noninvasive serum and imaging tests which can identify patients at risk for advanced fibrosis and liver-related complications. MASLD has been increasing in prevalence and disease burden but is underrecognized in primary care and endocrinology clinics. Multiple society guidelines, synthesized here, provide a framework for the initial approach in the diagnosis and evaluation of MASLD. Recent advances in pharmacologic treatment underline the importance of screening for patients who are at risk for advanced fibrosis as they are most likely to benefit from new drug classes, such as the liver-directed thyroid receptor agonist resmitemer.

Metabolic dysfunction-associated steatotic liver disease: Update and impact of new nomenclature on the American Association for the Study of Liver Diseases practice guidance on nonalcoholic fatty liver disease.

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This commentary discusses how clinicians and various stakeholders can utilize the recently published American Association for the Study of Liver Diseases nonalcoholic fatty liver disease (AASLD NAFLD) Practice Guidance in light of the change in the nomenclature to steatotic liver disease and its subcategories. The new terminologies explained in this commentary make it easier for the readers to interchangeably use metabolic dysfunction-associated steatotic liver disease (MASLD) in place of NAFLD and metabolic-dysfunction associated steatohepatitis (MASH) instead of nonalcoholic steatohepatitis (NASH), respectively, as they read the NAFLD Practice Guidance. The guidance document is relevant and can be utilized for the diagnosis, risk stratification, and management of patients with MASLD. This commentary serves as an accompanying article to the NAFLD Practice Guidance and helps its clinical application in the light of the new nomenclature.
